Faculty of Artificial Intelligence Engineering

Department of Informatics

Software Engineering and Information Systems



Language Learning Platform Using Games

Graduation Project 2 – Software Engineering and Information Systems

Prepared by

Noor Sheikha Ayat Mahmoud

Supervised by

Dr. Shadi Al-Bleidi

2026-2027

الملخص

Abstract

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Chapter (1):
Introduction

1.1 Project Overview and Background

The field of education continues to evolve as a result of advances in information technology and the widespread adoption of web applications and digital learning platforms. These developments have introduced modern educational approaches designed to make learning more flexible and interactive while considering learners' different needs and proficiency levels. Language learning is one of the fields that can benefit significantly from these technologies because it requires continuous practice and interaction with linguistic content in different contexts.

In this context, this project aims to develop an interactive language-learning platform based on digital games and educational activities. The platform combines educational content with interactive and motivational elements to help learners practice their language skills within an organized and accessible digital environment.

The platform provides several functions that support the learner's journey, including account registration and login, language selection, placement tests, educational games, and progress tracking. It also supports various learning activities, such as courses, interactive stories, listening challenges, and learned-vocabulary review.

1.2 Problem Statement and Significance

1.2.1 Problem Statement

The main problems addressed by this project can be summarized as follows:

1. Lack of learner engagement: Traditional language-learning methods may not provide sufficient interaction and motivation to maintain learners’ interest.
2. Limited practical learning opportunities: Learners may face difficulties practicing language skills through a variety of interactive activities.
3. Difficulty tracking learning progress: Learners may find it difficult to monitor their progress, identify their weaknesses, and evaluate their performance over time.
4. Lack of personalized learning: Educational activities may not always match learners’ different proficiency levels and individual needs.
5. Limited use of educational games: Some learning environments do not sufficiently integrate educational games into the language-learning process.
6. Difficulty retaining new vocabulary: Learners may struggle to remember and review previously learned words without appropriate revision activities.
7. Limited variety of learning activities: Learners need a variety of activities, such as interactive stories, listening challenges, quizzes, and educational games, to practice different language skills.

1.2.2 Project Significance

The significance of this project can be summarized in the following points:

1. Enhancing engagement and motivation: Providing an interactive learning environment based on educational games to make language learning more enjoyable and engaging.
2. Improving the learning experience: Offering a variety of educational activities that allow learners to practice languages through games, interactive stories, and listening challenges.
3. Supporting level-based learning: Enabling learners to determine their proficiency levels and access activities appropriate to their abilities and language skills.
4. Developing language skills: Supporting vocabulary practice, listening, and comprehension through diverse interactive activities.
5. Improving vocabulary retention: Allowing learners to review previously learned words, helping reinforce vocabulary retention.
6. Tracking learning progress: Enabling learners to monitor their progress, review their results, and identify their achievements throughout the learning process.
7. Providing continuous performance assessment: Using tests and educational activities to evaluate learners' proficiency and identify areas that require improvement.

1. 3 Proposed Solution

The proposed solution is to develop an interactive web-based language-learning platform that addresses challenges related to low learner engagement, limited variety in educational activities, and difficulties in tracking language-learning progress.

The platform will use modern web technologies and integrate educational games and interactive activities to provide an organized learning environment that combines practice, assessment, and performance tracking. The system will be developed using HTML, CSS, and JavaScript for the user interface, and PHP with the Laravel framework for back-end functionality.

The Model–View–Controller (MVC) architectural pattern will be adopted to organize the application components, while a database will be used to store user information, educational content, and activity results.

Furthermore, the Scrum framework, within the Agile methodology, will be used to organize development into short, iterative cycles, allowing requirements to be implemented, tested, and improved incrementally.

Proposed Solution Approach

The proposed solution will be implemented through the following steps:

1. Requirements Analysis and User Needs Identification: Identify the platform's core functions, define the requirements of learners and administrators, and establish the system structure based on these requirements.
2. System Architecture and Database Design: Define the platform components and their relationships, and design the database required to manage user accounts, languages, proficiency levels, activities, and results.
3. User Management and Placement Assessment: Implement registration, login, and profile management, along with placement tests that help learners identify their appropriate language proficiency levels.
4. Development of Interactive Learning Activities: Develop a variety of educational activities, including educational games, interactive quizzes, interactive stories, listening challenges, and vocabulary revision exercises, providing learners with different opportunities to practice language skills.
5. Implementation of Progress Tracking and Assessment: Record learner results, monitor learning progress, display performance and achievements, and provide reports that help users understand progress and identify areas requiring further practice.
6. Development of the Administration Dashboard: Create an administrative interface for managing users, languages, proficiency levels, courses, games, questions, and announcements, facilitating the organization and updating of educational content.
7. Application of Scrum for Development Management: Divide the project into short development cycles (Sprints), define tasks and priorities for each cycle, and implement system functions incrementally while reviewing results and improving the process at each stage.
8. System Testing and Evaluation: Test the core functions, verify data processing and component integration, evaluate usability, and resolve errors before delivering the final version.

1.4 Project Scope

The project focuses on designing and developing a web-based language-learning platform that uses educational games and interactive learning activities. The main system functions include:

1. User-account management, login, and profile updates.
2. Language selection and placement testing.
3. Displaying recorded language levels and reviewing previous test results.
4. Playing educational games and saving related results and progress.
5. Recommending games according to learner proficiency, where supported by the system.
6. Viewing educational courses and tracking course progress.
7. Providing interactive stories, listening challenges, and learned-vocabulary review.
8. Managing games, courses, languages, levels, and educational questions through an administrative dashboard.
9. Providing notifications, announcements, performance reports, and achievements, depending on the implemented functionality.

1.5 Project Objectives

1.5.1 Main Objectives

1. Develop an Interactive Language-Learning Platform: Create an educational environment that uses digital games and activities to provide an engaging and organized learning experience.
2. Enhance Learner Motivation: Incorporate motivational elements, such as challenges, results, and achievements, according to the functions supported by the system.
3. Personalize the Learning Experience: Direct learners toward activities appropriate to their proficiency levels using placement-test results and available learner data.
4. Support Flexible and Self-Paced Learning: Allow learners to access and practice educational activities at times suitable for their individual schedules.

1.5.2 Sub-Objectives

1. Design simple, clear, and user-friendly interfaces.
2. Improve system performance and minimize response times where possible.
3. Support use across compatible devices and web browsers.
4. Provide an organized visual design suitable for educational content.
5. Simplify account registration and login procedures.

1.6 State of the Art: Technologies Used in E-Learning Platform Development

A wide range of technologies and tools are used to develop e-learning websites and platforms. These technologies aim to provide interactive learning environments, enhance user experience, and efficiently manage educational content and data. The main categories include the following:

- ✓ Front-End Technologies :HTML5 ,CSS3 ,JavaScript ,React.js ,Vue.js ,Angular
- ✓ Back-End Technologies :Laravel ,Django ,ASP.NET Core ,ava Spring Boot
- ✓ Interface Design and Development Tools: Figma ,Adobe XD ,Git ,GitHub ,Visual Studio Code

Artificial Intelligence Technologies

1. Natural Language Processing (NLP)
2. Machine Learning (ML)
3. Educational Recommendation Systems
4. Educational Chatbots
5. Adaptive Learning Systems

E-Learning Standards and Technologies

1. Learning Management Systems (LMS)
2. Sharable Content Object Reference Model (SCORM)
3. Experience API (xAPI)
4. Learning Tools Interoperability (LTI)